

Your Cancer Summary Report

Michelle Thompson

Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma

Report Date: May 06, 2026

⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

At a Glance

Your Diagnosis

Breast Cancer - Invasive Ductal Carcinoma (IDC)

Stage: Stage IIA | **Grade:** Grade 2

You have an early-stage breast cancer that responds to hormones like estrogen and progesterone. This type of breast cancer, called hormone receptor-positive, has very effective targeted treatments and a favorable outlook.

✓ Positive Factors

- Hormone receptor-positive (ER+/PR+) — responds well to hormone-blocking therapy
- HER2-negative confirmed by spatial transcriptomics — avoids HER2 toxicity
- BRCA2 mutation makes you eligible for PARP inhibitor therapy (olaparib)
- PIK3CA H1047R mutation opens eligibility for alpelisib + fulvestrant
- Lab markers CEA (2.1) and CA 15-3 (18) are within normal range
- Stage IIA — localized disease with favorable prognosis

⚠ Challenges to Consider

- BRCA2 germline mutation has family implications — genetic counseling recommended

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

- CCND1 amplification associated with CDK4/6 inhibitor sensitivity

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

PIK3CA

Pathogenic

H1047R missense mutation (VAF 0.42)

A change in the PIK3CA gene that makes cancer cells grow faster. This change is common in hormone receptor-positive breast cancer and can be targeted with a specific drug called alpelisib.

Treatment Option: Alpelisib + fulvestrant (FDA Approved — SOLAR-1 trial)

BRCA2

Pathogenic

Frameshift variant — c.5946delT (VAF 0.50)

A significant change in the BRCA2 gene, which normally helps repair DNA. This change means your cancer cells have trouble repairing DNA damage, which makes them sensitive to a class of drugs called PARP inhibitors.

Treatment Option: Olaparib / Niraparib — PARP inhibitor (FDA Approved). HRD score 35 (below threshold) but BRCA2 mutation independently confers PARPi eligibility.

GATA3

Likely Pathogenic

Frameshift variant (VAF 0.31)

A change in the GATA3 gene, a key regulator of luminal breast cancer identity. This is common in ER-positive breast cancers and may influence how well hormone therapy works.

Treatment Option: Marker of luminal A subtype; monitor for

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

MAP3K1

VUS

p.R264H missense (VAF 0.35)

A change in the MAP3K1 gene of uncertain importance. It is being tracked but does not currently change your treatment plan.

ERBB2 (HER2)

Wild-type

Copy number neutral (cn=2) — HER2-negative confirmed

Your cancer does not have extra copies of the HER2 gene — confirmed by both genomic and spatial transcriptomics analysis. This means HER2-targeted drugs like trastuzumab are not indicated.

Treatment Option: No HER2-targeted therapy indicated

CCND1

Pathogenic

Copy number gain (cn=6, log2 1.62)

Extra copies of the CCND1 gene, which drives cell cycle progression. This amplification is associated with CDK4/6 inhibitor sensitivity.

Treatment Option: Palbociclib / ribociclib / abemaciclib (CDK4/6 inhibitors, FDA Approved)

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

Luminal A profile confirmed spatially. Patient context analysis suggests ESR1/PGR/GATA3/FOXA1 elevated in tumor core vs stroma. ERBB2 low — HER2-negative validated spatially. Spatial CSV files (PAT002_expression.csv, PAT002_coordinates.csv, PAT002_regions.csv) not yet generated for full parsing — context set for 900 spots across 5 regions.

Immune Status: • Mixed immune activity - Variable immune response across the tumor

Mapping of gene activity across your tumor tissue confirms the hormone-positive, HER2-negative character of your cancer. Immune cells are present in the tumor area, which is a modestly positive sign.

Key Patterns Observed

- Luminal A expression profile (ESR1, PGR, GATA3, FOXA1 elevated)
- ERBB2/HER2 spatially low — confirms HER2-negative status
- MKI67 moderate — Grade 2 proliferation consistent
- 5 regions: tumor_core, adipose, stroma, immune_infiltrate, normal_epithelium
- Immune infiltrate region present — mixed TME classification

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen (Continuation)

targeted

FDA Approved

Expected Response: 50-60%

Tamoxifen blocks estrogen from fueling breast cancer cells. You are already taking 20mg daily. This is the standard of care for premenopausal ER-positive breast cancer.

Why this may help you: ER+/PR+ hormone receptor status — confirmed by genomics and spatial ESR1/PGR expression

Common side effects: Hot flashes, Mood changes, Blood clot risk (rare), Uterine changes with long-term use

Alpelisib + Fulvestrant

targeted

FDA Approved

Expected Response: 35-40%

Alpelisib is a pill that blocks the PIK3CA pathway. It is combined with fulvestrant, a hormone-blocking injection. This combination specifically targets the PIK3CA H1047R mutation found in your tumor.

Why this may help you: PIK3CA H1047R mutation (VAF 0.42) — FDA Approved per SOLAR-1 trial

Common side effects: High blood sugar (manage with diet/medication), Diarrhea, Rash, Fatigue, Nausea

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

Palbociclib (CDK4/6 inhibitor)

targeted

FDA Approved

Expected Response: 55-60%

Palbociclib is a pill that slows cancer cell division by blocking a cell cycle protein (CDK4/6). It is typically combined with hormone therapy.

Why this may help you: CCND1 amplification (cn=6) — strong CDK4/6 inhibitor sensitivity signal

Common side effects: Low white blood cell count (requires monitoring), Fatigue, Nausea, Hair thinning

Olaparib (PARP Inhibitor)

targeted

FDA Approved

Expected Response: 59% (OlympiAD trial)

Olaparib exploits the DNA repair defect caused by your BRCA2 mutation. It works by blocking another DNA repair pathway, causing cancer cells to die while sparing normal cells.

Why this may help you: Confirmed pathogenic BRCA2 frameshift mutation — independently eligible regardless of HRD score of 35

Common side effects: Nausea, Fatigue, Anemia (low red blood cells), Low platelet count

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 3 months CA 15-3 tumor marker
Monitor cancer activity — baseline CA 15-3 is 18 (normal)

Every 3 months CEA tumor marker
Additional marker monitoring — baseline CEA is 2.1 (normal)

Every 6 months Breast MRI or mammogram
Imaging surveillance for local recurrence or new lesions

Before starting alpelisib, then monthly Blood glucose (fasting)
Alpelisib can raise blood sugar — monitoring is required

Every 28 days on palbociclib Complete blood count (CBC)
CDK4/6 inhibitors can reduce white blood cells

Annually Bone density scan (DEXA)
Long-term tamoxifen use can affect bone density

⚠ When to Call Your Doctor

Contact your care team right away if you experience:

- New breast lump or swelling in the armpit
- Bone pain or back pain lasting more than 2 weeks
- Shortness of breath or chest pain
- Severe diarrhea or skin rash (on alpelisib)
- Unusual bruising or prolonged bleeding
- Fever over 38°C / 100.4°F (especially on CDK4/6 inhibitor)

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

Questions? Your oncology care team or nurse navigator — available weekdays; emergency line 24/7 for urgent symptoms

For Your Family

Your BRCA2 mutation is hereditary, meaning it may be present in blood relatives. First-degree relatives (parents, siblings, children) have a 50% chance of carrying this mutation. BRCA2 mutations increase risk of breast, ovarian, and pancreatic cancers.

Recommended Steps for Family Members

- Discuss BRCA2 result with first-degree relatives (siblings, children, parents)
- Refer all interested relatives to genetic counseling
- Daughters should begin breast cancer screening earlier than standard guidelines

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Important Notice

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.

Data Sources: mockepic, fgbio, genomic-results, spatialtools, multiomics, patient-report

Report Version: 1.0

Generated: 2026-05-06 20:48:18

This AI-generated summary is based on SYNTHETIC DATA and must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All findings are from SYNTHETIC_DATA mode — not for clinical use.